QP Code:

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION AUGUST 2025
THIRD SEMESTER
Discipline Specific Elective (DSE) - MG3DSEPHY204
INTRODUCTION TO MATERIALS SCIENCE- PRACTICAL EXAMINATION
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks:** 35

Record: 5 Marks

1. From the given X-ray diffraction (XRD) pattern of a crystalline sample, estimate the Miller indices (hkl) of the observed diffraction peaks. Include all steps, calculations, and a final table of indices. [An]/[CO7]

2. Draw a cube and neatly sketch all the crystallographic planes and directions corresponding to the primary Miller indices. Label each plane and direction clearly with proper notation.

[An]/[CO7]

3. Using the provided XRD data of a simple cubic crystal, determine the lattice constant (a) of the material. Show all relevant calculations and interpretations.
[An]/[CO7]

4. Construct a 3D model of a simple cubic unit cell. Clearly indicate a specific crystallographic direction and a plane within the unit cell using standard notation. [An]/[CO7]

5. Construct a 3D model of an orthorhombic unit cell. Identify and mark one specific crystallographic direction and one plane as per instructions. [An]/[CO7]

6. Construct a 3D model of a monoclinic unit cell. Clearly represent a given crystallographic direction and plane within the cell, and label them appropriately. [An]/[CO7]

7. Interpret the given binary phase diagram and answer the following:

- Identify the phases present at specific temperatures and compositions.
- Mark eutectic/eutectoid/peritectic points, if any.
- Explain the phase transformations occurring during cooling. [An]/[CO7]

8. Using the standard experimental setup (bending or elongation method), determine the modulus of elasticity of the provided metal/alloy sample. Include all measurements, formulas used, calculations, and final result with proper units. [An]/[CO7]